Maulshree Jatrana

LinkedIn | GitHub | Portfolio

Education

Chitkara University Institute of Engineering and Technology Aug 2023 – Jul 2027
B.E. in Computer Science and Engineering (Artificial Intelligence) *CGPA: 9.89 / 10*
Class XII: 92.4% | CBSE, KV FAD Dappar, 2022 Class X: 94.4% | CBSE, KV FAD Dappar, 2020

Technical Skills

- **Languages:** C++, Java, Python, SQL, JavaScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems (Unix/Linux), DBMS, SDLC
- **Backend & Systems:** REST API Design, Node.js, Express.js, JWT Authentication, Scalable Distributed Systems
- **AI/ML:** CNNs, PyTorch, Dataset Preprocessing, Model Training, Model Evaluation, Computer Vision
- **Agentic AI:** Hybrid RAG, FAISS, BM25, Cross-Encoder Reranking, LangChain, LangGraph
- **Databases:** MongoDB, PostgreSQL, MySQL, SQLite
- **Tools & Cloud:** Git, Docker, Linux/Unix, Postman, GCP, AWS, Nutanix (NCA-6)

Experience

HighRadius Jul 2026 – Present
Forward Deployed Engineer Intern
- Building **AI agents** for HighRadius's **Deductions Management (DMS)** product, automating how customer **short-payments** are investigated and resolved within the **Order-to-Cash** cycle.
- Completed intensive training on the **Order-to-Cash (O2C)** cycle; work as a **forward deployed engineer**, translating finance-operations requirements into working AI workflows.

ServiceNow Jun 2025 – Aug 2025
Software Development Engineer Intern *Remote*
- Engineered **automated enterprise workflows** across **development and testing stages of SDLC**, translating complex business requirements into scalable production-grade solutions.
- Designed **automation pipelines** and **reporting systems** using **low-code architecture**, improving **integration capability** and **system performance** across large-scale workflows.

Microsoft – Code Without Barriers Scholar Apr 2025 – Jun 2025
Software Development Program *Remote*
- Optimized **relational database schemas** handling **50K+ daily transactions**, improving **query performance by 25%** using advanced indexing and query tuning.
- Designed **advanced SQL queries** (JOINs, CTEs, window functions), reducing **latency from 45s to 12s** and improving **data scalability**.

Projects

Chargeback Shield – AI Risk Manager for Payment Disputes [GitHub | Live Demo] Sep 2026
Python, FastAPI, scikit-learn, Docker, Render
- Built a two-stage payment-risk system: a **HistGradientBoosting** model predicts **dispute risk pre-authorization**, and a deterministic policy engine assembles **reason-code-specific evidence** and recommends fight/concede/review once a dispute is filed.
- Engineered **leakage-safe, time-ordered features** and reached **40% precision / 40.6% recall** on a held-out test set, a **~60% lift** over a naive baseline; shipped with **Docker**, **CI (23 tests)**, append-only audit logging, and a live production deployment.

Accreditation Copilot – Hybrid Retrieval System [GitHub] 2026
Python, C++, PyTorch, FAISS, BM25, SQLite
- Built a production-grade AI retrieval system using **FAISS (dense) + BM25 (sparse)** with **RRF fusion** and **cross-encoder reranking**; implemented **C++ indexing** for high-throughput, low-latency performance.
- Achieved **P@1: 80%, P@5: 100%, ~90% MRR**, and **~625ms latency**; improved context enrichment by **2.8x**.

SyncUp – AI Collaboration Platform [GitHub | Demo] Oct 2025
ReactJS, Node.js, Express.js, MongoDB, LangGraph
- Developed **multi-agent workflows** using **LangGraph** with contextual memory and LLM orchestration for intelligent team formation.
- Built modular **REST APIs** with **JWT authentication**, validation, and session management; tested and validated system reliability using Postman.

MindForge – Deepfake Image Detection System [GitHub | Demo] Mar 2025
Python, PyTorch, TensorFlow, OpenCV
- Designed and trained a **CNN-based model** achieving **96% accuracy** using data augmentation and optimized training pipelines.
- Built an end-to-end **ML pipeline** with **REST API inference endpoints** enabling real-time scalable deepfake detection.

Achievements & Certifications

- **Flipkart GRID 7.0** – National Semi-Finalist, ranked in the **Top 1%** among **200,000+ participants** in a national engineering competition.
- **Hackfest 2025** – Top 10 out of 120 teams, evaluated on innovation, scalability, and practical applicability.
- **Published Research (First Author) –** View Paper, focused on AI-based ocean water management (2024).
- **Nutanix Certified Associate (NCA 6)** – Certification in Multicloud Infrastructure, Cloud Architecture, and Virtualization (Feb 2026).